

deepSSF Data Prep

Scott Forrest

2026-05-26

Here we prepare the data for fitting a deepSSF model. We load the GPS data and tidy it, create a trajectory object containing a series of steps and read in the environmental layers. For each observed step we crop out local subsets of the environmental layers, centred on the step's location. For every step we will then have the surrounding environmental information stored as a stack of small rasters, with the 'target' of what we're trying to predict (the actual location of the next step) as a raster layer as well, with all cells being 0 except the observed next location, which is 1. These, along with temporal covariates and the previous bearing, will be the input data for the deepSSF model.

Table of contents

Loading packages	2
Import data	2
Tidy data	3
Create a subset of the data to export for testing the deepSSF package	4
Setup trajectory	4
Plot the data coloured by time	5
Plot a timeline of each individual	5
Number of location per day	6
Generating the data to fit a deepSSF model	7
Create a steps object to check the step lengths	7
Plot step lengths and set the buffer distance	8
Prepare and export data for single layer saving (Python script)	10
Save the data to a csv file	11
Read in the environmental covariates	12
Create some layers for testing the deepSSF package	15
Set up the spatial extent of the local covariates	17
Loop over each individual and save the local rasters	18
Check the outputs	22

Plot a subset of the local covariates	24
To save the object with all of the local covariates	36
To save some plots	36
Plot the step lengths and turning angles	37

Loading packages

```
library(tidyverse)
packages <- c("amt", "sf", "terra", "beepR", "tictoc")
walk(packages, require, character.only = T)
```

Import data

```
buffalo <- read_csv("data/buffalo.csv")
```

New names:

Rows: 133161 Columns: 11

-- Column specification

```
----- Delimiter: "," chr
(3): node, dates, DateTime dbl (7): ...1, lat, lon, height, accuracy, heading,
speed dtm (1): timestamp
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```
head(buffalo)
```

A tibble: 6 x 11

	...1 timestamp	lat	lon	height	accuracy	heading	speed	node
	<dbl> <dtm>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	169 2018-07-25 00:04:02	-12.3	134.	91.3	977	0	0	2005
2	170 2018-07-25 01:04:23	-12.3	134.	92.1	704	77.0	0.034	2005
3	171 2018-07-25 02:04:39	-12.3	134.	95.2	1105	129.	0.008	2005
4	172 2018-07-25 03:04:17	-12.3	134.	104.	865	0	0.013	2005
5	173 2018-07-25 04:04:39	-12.3	134.	85.5	711	231.	0.01	2005
6	174 2018-07-25 05:04:27	-12.3	134.	95.8	780	207.	0.012	2005

i 2 more variables: dates <chr>, DateTime <chr>

Tidy data

```
# remove individuals that have poor data quality or less than about 3 months of data.
# The "2014.GPS_COMPACT copy.csv" string is a duplicate of ID 2024, so we also exclude it
buffalo <- buffalo %>% filter(!node %in% c("2014.GPS_COMPACT copy.csv",
                                           # 2005, 2014, 2018, 2021, 2022, 2024,
                                           2029, 2043, 2265, 2284, 2346, 2354))

# arrange by time and exclude any duplicate timestamps (within the same individual's data)
buffalo <- buffalo %>%
  group_by(node) %>%
  arrange(DateTime, .by_group = T) %>%
  distinct(DateTime, .keep_all = T) %>%
  arrange(node) %>%
  mutate(ID = node)

# keep only the relevant columns
buffalo_clean <- buffalo[, c(12, 2, 4, 3)]

# rename the columns
colnames(buffalo_clean) <- c("id", "time", "lon", "lat")

# convert the time to the correct timezone
attr(buffalo_clean$time, "tzone") <- "Australia/Queensland"
head(buffalo_clean)
```

```
# A tibble: 6 x 4
  id    time                lon  lat
<chr> <dtm>                <dbl> <dbl>
1 2005 2019-01-01 11:01:14  134. -12.3
2 2005 2019-01-01 21:00:59  134. -12.3
3 2005 2019-01-01 22:01:15  134. -12.3
4 2005 2019-01-01 23:01:28  134. -12.3
5 2005 2019-01-02 00:01:12  134. -12.3
6 2005 2019-01-02 03:00:36  134. -12.3
```

```
# check timezone
tz(buffalo_clean$time)
```

```
[1] "Australia/Queensland"
```

```
# create a vector of the unique ids for indexing later
buffalo_ids <- unique(buffalo_clean$id)
```

Create a subset of the data to export for testing the deepSSF package

We'll select the ID 2005, as that was used for training the model in the MEE paper

```
buffalo_clean_2005 <- buffalo_clean %>% filter(id == 2005)

# convert to sf object and transform to the same projection as the covariates (GDA94 / Geoscience Australia Lambert)
buffalo_clean_2005_sf <- buffalo_clean_2005 %>%
  st_as_sf(coords = c("lon", "lat"), crs = 4326) %>%
  st_transform(crs = 3112)

# add the coordinates back to the data frame
buffalo_clean_2005_coords <- buffalo_clean_2005_sf %>%
  st_coordinates() %>%
  as.data.frame() %>%
  rename(x = X, y = Y)

buffalo_clean_2005_proj <- buffalo_clean_2005 %>%
  bind_cols(buffalo_clean_2005_coords) %>%
  dplyr::select(id, time, x, y)

buffalo_clean_2005_proj %>% write_csv("data/buffalo_djelk_id2005.csv")
```

Setup trajectory

Use the amt package to create a trajectory object from the cleaned data.

```
buffalo_all <- buffalo_clean %>% mk_track(id = id,
                                          lon,
                                          lat,
                                          time,
                                          all_cols = T,
                                          crs = 4326) %>%
  # Transformation to GDA94 / Geoscience Australia Lambert (https://epsg.io/3112)
  transform_coords(crs_to = 3112, crs_from = 4326) %>% arrange(id)

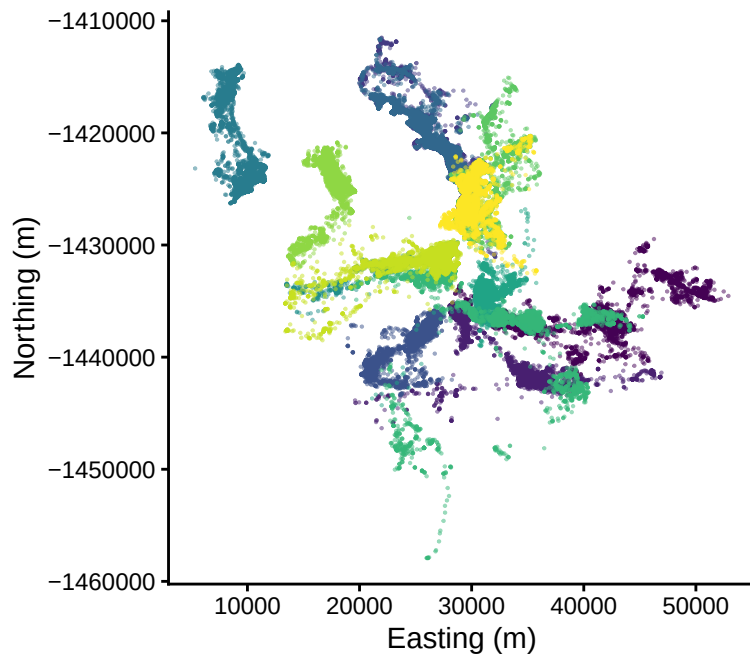
# Number of IDs and locations
buffalo_all %>%
  summarise(n_ids = n_distinct(id),
            n_locations = n())
```



```
# A tibble: 1 x 2
  n_ids n_locations
  <int>   <int>
1     13    107554
```

Plot the data coloured by time

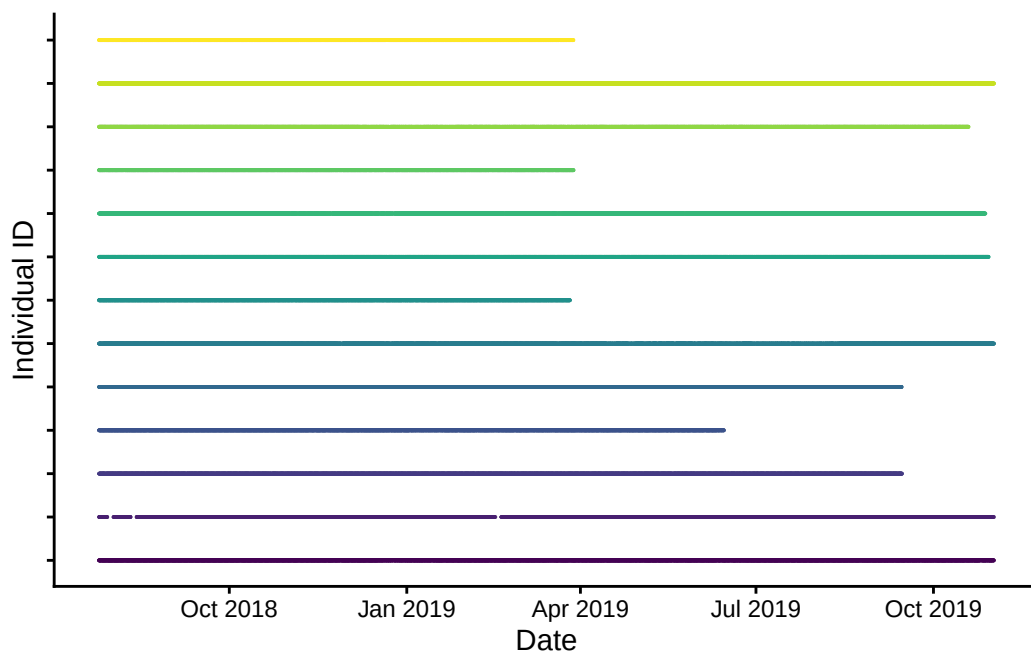
```
buffalo_all %>%
  ggplot(aes(x = x_, y = y_, colour = id)) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "none")
```



```
ggsave("outputs/data_prep/buffalo_djelk_map.png",
  width = 150, height = 150, units = "mm", dpi = 1000)
```

Plot a timeline of each individual

```
buffalo_all %>%
  ggplot(aes(x = t_, y = as.factor(id), colour = as.factor(id))) +
  geom_point(alpha = 0.5, size = 0.01) +
  scale_x_datetime("Date") +
  scale_y_discrete("Individual ID") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "none",
        axis.text.y = element_blank())
```

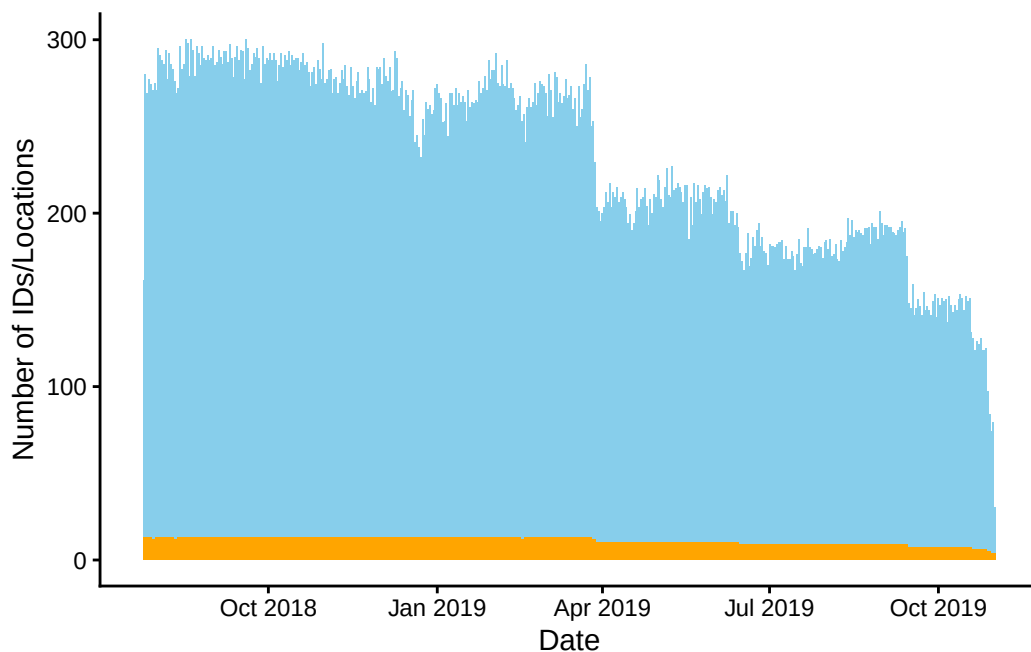


```
ggsave("outputs/data_prep/buffalo_djelk_timeline.png",
       width = 150, height = 150, units = "mm", dpi = 600)
```

Number of location per day

```
buffalo_daily_locations <- buffalo_all %>%
  mutate(date = lubridate::date(t_)) %>%
  dplyr::group_by(date) %>%
  summarise(
    n_locations = n(),
    n_individuals = n_distinct(id)
  )
```

```
buffalo_daily_locations %>%
  ggplot() +
    geom_col(aes(x = date, y = n_locations),
              fill = "skyblue") +
    geom_col(aes(x = date, y = n_individuals),
              fill = "orange") +
    scale_x_date("Date") +
    scale_y_continuous("Number of IDs/Locations") +
    scale_colour_viridis_d() +
    theme_classic() +
    theme(legend.position = "none")
```



```
ggsave("outputs/data_prep/buffalo_djelk_daily_locations.png",
        width = 150, height = 80, units = "mm", dpi = 600)
```

Generating the data to fit a deepSSF model

Create a steps object to check the step lengths

```
# nest the data by individual
buffalo_all_nested <- buffalo_all %>% arrange(id) %>% nest(data = "-id")
```

```

buffalo_all_nested_steps <- buffalo_all_nested %>%
  mutate(steps = map(data, function(x)
    x %>%
      # track_resample(rate = hours(1), tolerance = minutes(10)) %>%
      steps()))

# unnest the data after creating 'steps' objects
buffalo_all_steps <- buffalo_all_nested_steps %>%
  amt::select(id, steps) %>%
  amt::unnest(cols = steps)

head(buffalo_all_steps)

```

```

# A tibble: 6 x 11
  id      x1_    x2_      y1_      y2_    sl_ direction_p    ta_
* <chr> <dbl> <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
1 2005  41941. 41969. -1435875. -1435671. 206.      1.43    NA
2 2005  41969. 41922. -1435671. -1435654.  50.7     2.80    1.37
3 2005  41922. 41779. -1435654. -1435601. 152.     2.78   -0.0214
4 2005  41779. 41841. -1435601. -1435635.  70.7    -0.507    2.99
5 2005  41841. 41655. -1435635. -1435604. 188.     2.98   -2.80
6 2005  41655. 41619. -1435604. -1435608.  37.1    -3.02    0.285
# i 3 more variables: t1_ <dtm>, t2_ <dtm>, dt_ <drtn>

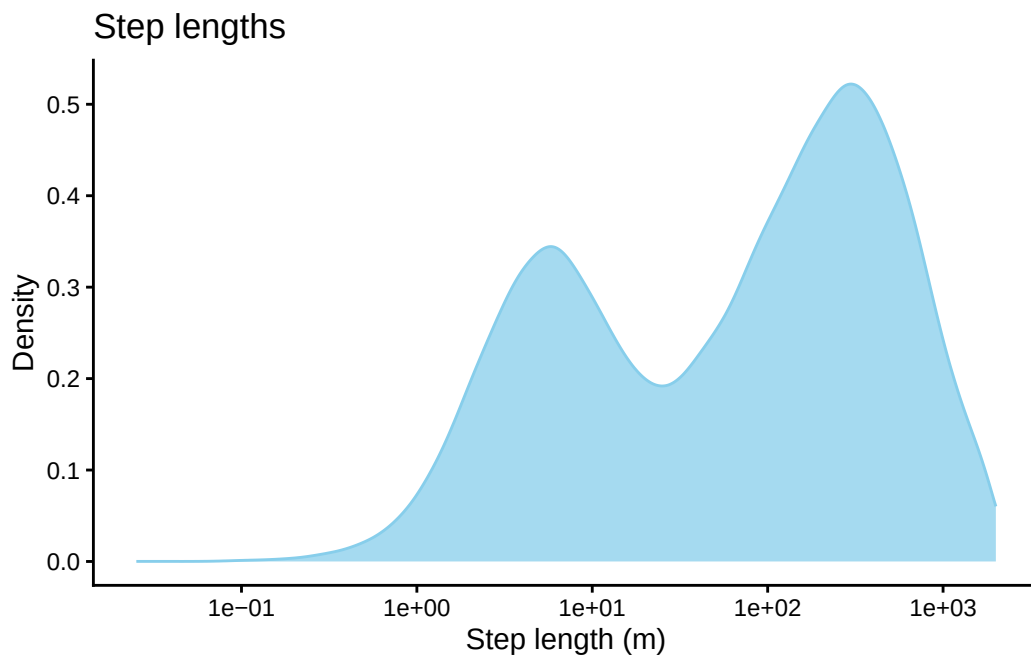
```

Plot step lengths and set the buffer distance

```

# plot step lengths
buffalo_all_steps %>%
  filter(sl_ < 2000) %>%
  ggplot() +
  geom_density(aes(x = sl_), #, bins = 50
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  # scale_x_continuous("Step length (m)") + #, limits = c(-25, 1250)
  scale_x_log10("Step length (m)") + #, limits = c(-25, 1250)
  scale_y_continuous("Density") +
  ggtitle("Step lengths") +
  theme_classic()

```



```
# proportion of steps less than the buffer distance
# the distance should be a multiple of the resolution
distance <- 1250
res <- 25

# check if multiple of the resolution?
distance %% res == 0
```

```
[1] TRUE
```

```
# buffer distance (add the resolution of the cell/2 as we want a centre cell)
buffer <- distance + (res/2)

# calculate the number of cells in each axis
nxn_cells <- buffer*2/res
print(paste0("Number of cells in each axis: ", nxn_cells))
```

```
[1] "Number of cells in each axis: 101"
```

```
# calculate proportion
prop_steps <- buffalo_all_steps %>% filter(x2_ - x1_ > -buffer &
                                           x2_ - x1_ < buffer &
                                           y2_ - y1_ > -buffer &
                                           y2_ - y1_ < buffer) %>%
```

```
nrow() / buffalo_all_steps %>% nrow()

print(paste0("Proportion of steps less than ", buffer, "m in x or y direction: ", round(prop
```

```
[1] "Proportion of steps less than 1262.5m in x or y direction: 0.9637"
```

```
# maximum distance at corner
max_dist <- sqrt(buffer^2 + buffer^2)
print(paste0("Maximum distance at corner: ", round(max_dist, 2)))
```

```
[1] "Maximum distance at corner: 1785.44"
```

Prepare and export data for single layer saving (Python script)

We also have another script that saves all local layers as individual npy arrays, which works better when there are many steps per id (to prevent large tif files), and can make things a bit easier for training models with many ids.

```
buffalo_all_steps_save <- buffalo_all_steps %>% mutate(

  dt_minute = as.numeric(round(difftime(t2_, t1_, units = "mins"), 0)),
  dt_hour = as.numeric(round(difftime(t2_, t1_, units = "hours"), 2)),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t1 = round(lubridate::hour(t1_) + lubridate::minute(t1_)/60,2),
  yday_t1 = round(lubridate::yday(t1_) + lubridate::hour(t1_)/24,2),

  hour_t1_sin1 = sin(2*pi*hour_t1/24),
  hour_t1_cos1 = cos(2*pi*hour_t1/24),
  hour_t1_sin2 = sin(4*pi*hour_t1/24),
  hour_t1_cos2 = cos(4*pi*hour_t1/24),

  yday_t1_sin1 = sin(2*pi*yday_t1/365.25),
  yday_t1_cos1 = cos(2*pi*yday_t1/365.25),
  yday_t1_sin2 = sin(4*pi*yday_t1/365.25),
  yday_t1_cos2 = cos(4*pi*yday_t1/365.25),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t2 = round(lubridate::hour(t2_) + lubridate::minute(t2_)/60,2),
  yday_t2 = round(lubridate::yday(t2_) + lubridate::hour(t2_)/24,2),

  bearing = direction_p,
```

```

    bearing_sin = sin(bearing),
    bearing_cos = cos(bearing)
)

```

```

buffalo_all_steps_save

```

```

# A tibble: 107,541 x 28

```

```

  id      x1_      x2_      y1_      y2_      sl_ direction_p      ta_
* <chr> <dbl> <dbl>      <dbl>      <dbl> <dbl>      <dbl> <dbl>
1 2005  41941.  41969. -1435875. -1435671. 206.        1.43  NA
2 2005  41969.  41922. -1435671. -1435654.  50.7        2.80  1.37
3 2005  41922.  41779. -1435654. -1435601. 152.        2.78 -0.0214
4 2005  41779.  41841. -1435601. -1435635.  70.7       -0.507  2.99
5 2005  41841.  41655. -1435635. -1435604. 188.        2.98 -2.80
6 2005  41655.  41619. -1435604. -1435608.  37.1       -3.02  0.285
7 2005  41619.  41688. -1435608. -1436126. 522.       -1.44  1.58
8 2005  41688.  41684. -1436126. -1436119.  8.13        2.10 -2.75
9 2005  41684.  41675. -1436119. -1436119.  9.75       -3.13  1.05
10 2005  41675.  41425. -1436119. -1435938. 308.        2.52 -0.635
# i 107,531 more rows
# i 20 more variables: t1_ <dtm>, t2_ <dtm>, dt_ <drtn>, dt_minute <dbl>,
#   dt_hour <dbl>, hour_t1 <dbl>, yday_t1 <dbl>, hour_t1_sin1 <dbl>,
#   hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>, hour_t1_cos2 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, hour_t2 <dbl>, yday_t2 <dbl>, bearing <dbl>,
#   bearing_sin <dbl>, bearing_cos <dbl>

```

Save the data to a csv file

```

# remove steps that fall outside of the local spatial extent
buffalo_all_steps_save <- buffalo_all_steps_save %>%
  filter(x2_ - x1_ > -distance &
         x2_ - x1_ < distance &
         y2_ - y1_ > -distance &
         y2_ - y1_ < distance) %>%
  drop_na(ta_)

buffalo_all_steps_save

```

```

# A tibble: 103,558 x 28

```

```

  id      x1_      x2_      y1_      y2_      sl_ direction_p      ta_

```

```

      <chr> <dbl> <dbl>      <dbl>      <dbl> <dbl>      <dbl> <dbl>
1 2005  41969. 41922. -1435671. -1435654.  50.7      2.80  1.37
2 2005  41922. 41779. -1435654. -1435601. 152.      2.78 -0.0214
3 2005  41779. 41841. -1435601. -1435635.  70.7     -0.507  2.99
4 2005  41841. 41655. -1435635. -1435604. 188.      2.98 -2.80
5 2005  41655. 41619. -1435604. -1435608.  37.1     -3.02  0.285
6 2005  41619. 41688. -1435608. -1436126. 522.     -1.44  1.58
7 2005  41688. 41684. -1436126. -1436119.   8.13     2.10 -2.75
8 2005  41684. 41675. -1436119. -1436119.   9.75     -3.13  1.05
9 2005  41675. 41425. -1436119. -1435938. 308.      2.52 -0.635
10 2005  41425. 40563. -1435938. -1435841. 867.      3.03  0.513
# i 103,548 more rows
# i 20 more variables: t1_ <dtm>, t2_ <dtm>, dt_ <drtn>, dt_minute <dbl>,
#   dt_hour <dbl>, hour_t1 <dbl>, yday_t1 <dbl>, hour_t1_sin1 <dbl>,
#   hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>, hour_t1_cos2 <dbl>,
#   yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
#   yday_t1_cos2 <dbl>, hour_t2 <dbl>, yday_t2 <dbl>, bearing <dbl>,
#   bearing_sin <dbl>, bearing_cos <dbl>

```

```

buffalo_all_steps_save %>% write_csv(paste0("data/buffalo_djelk_all_steps_nxn", nxn_cells,".

```

In our case over 96% of the steps were less than a distance of 1262.5m in either the x or y direction, which would mean that including 101 x 101 cells (which would be 2525 m x 2525 m) would include at least 96% of the steps. As the distance is further to the corner (but still less than the buffer distance in the x and y directions), there will be step lengths longer than the buffer distance (up to 1785 m).

Read in the environmental covariates

```

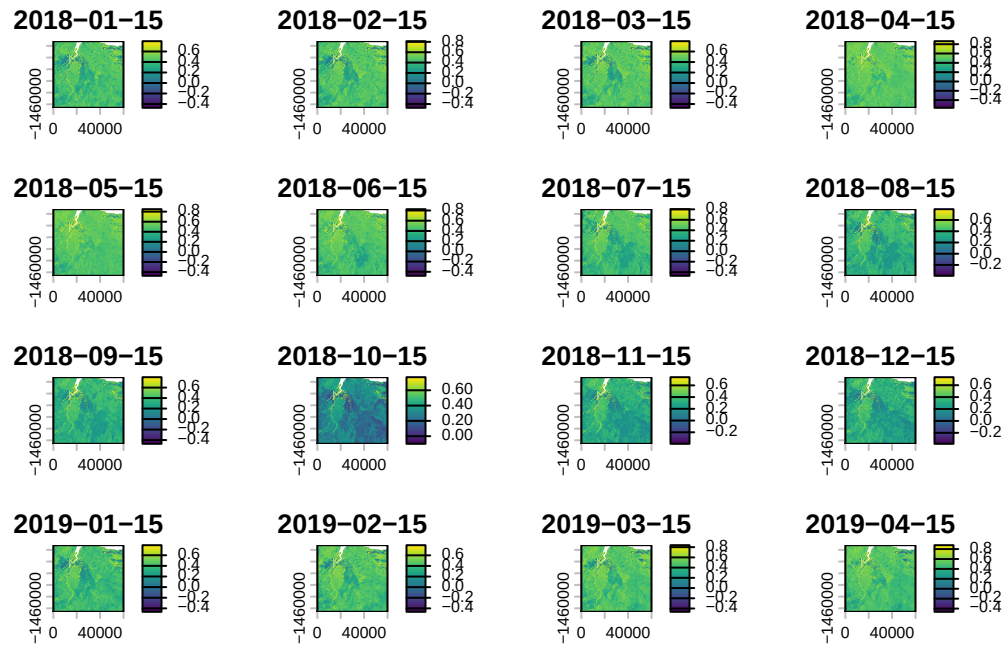
ndvi_projected <- rast("mapping/cropped rasters/ndvi_GEE_projected_watermask20230207.tif")
terra::time(ndvi_projected) <- as.POSIXct(lubridate::ymd("2018-01-15") + months(0:23))
slope <- rast("mapping/cropped rasters/slope_raster.tif")
veg_herby <- rast("mapping/cropped rasters/veg_herby.tif")
canopy_cover <- rast("mapping/cropped rasters/canopy_cover.tif")

# change the names (these will become the column names when extracting
# covariate values at the used and random steps)
names(ndvi_projected) <- rep("ndvi", terra::nlyr(ndvi_projected))
names(slope) <- "slope"
names(veg_herby) <- "veg_herby"
names(canopy_cover) <- "canopy_cover"

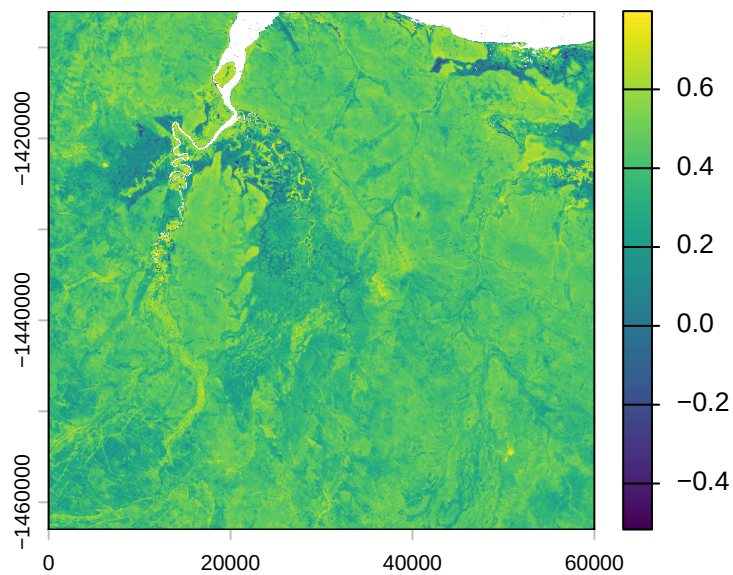
```



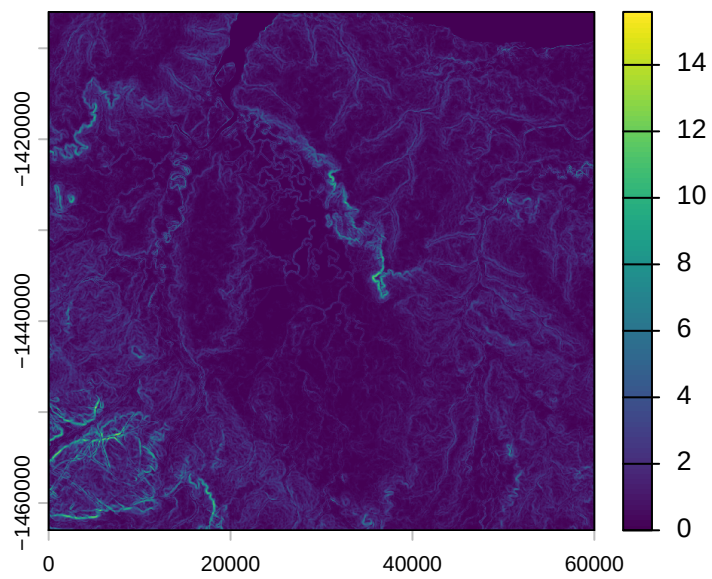
```
# to plot the rasters
plot(ndvi_projected)
```



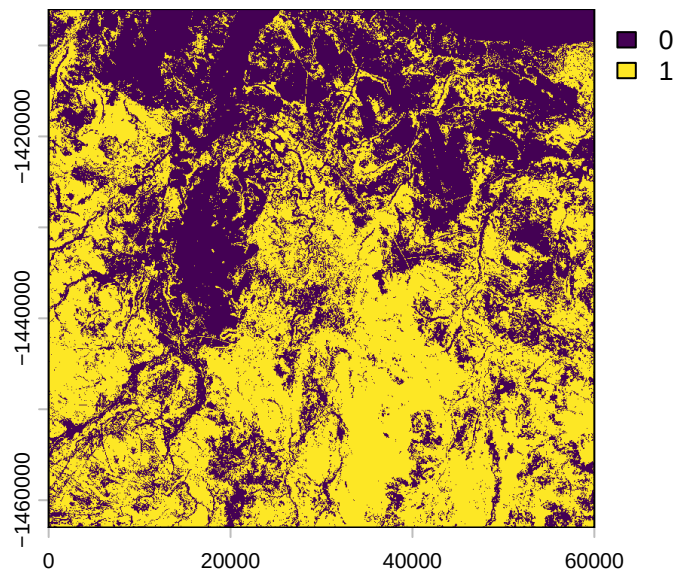
```
plot(ndvi_projected[["1"]])
```



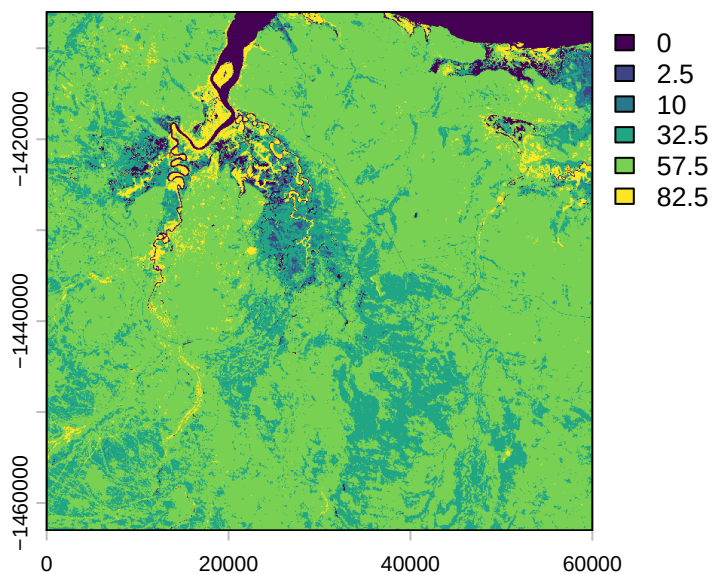
```
plot(slope)
```



```
plot(veg_herby)
```



```
plot(canopy_cover)
```

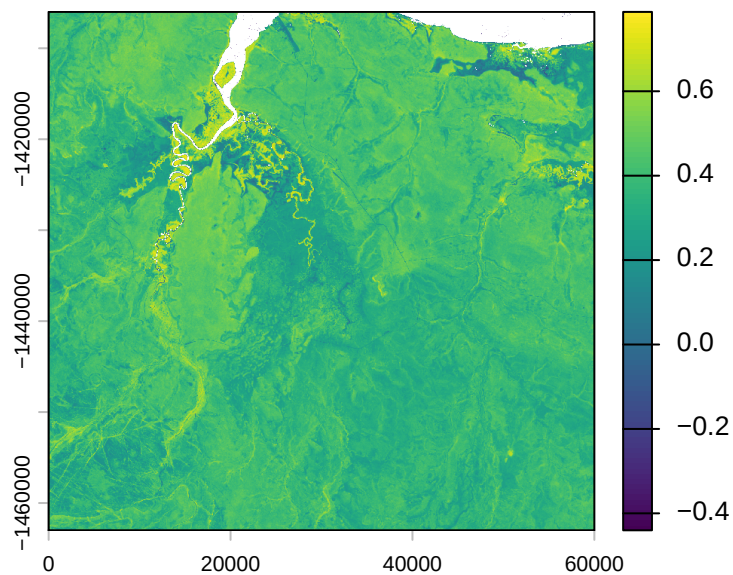


```
# get the resolution from the covariates  
res <- terra::res(ndvi_projected)[1]
```

Create some layers for testing the deepSSF package

These will be cropped to the extent of the ID 2005 data. We will also take an average of the NDVI values for simplicity.

```
buffer_distance <- 2500 # add a generous buffer distance to ensure we capture all steps for  
  
ndvi_average <- terra::app(ndvi_projected, mean)  
plot(ndvi_average)
```

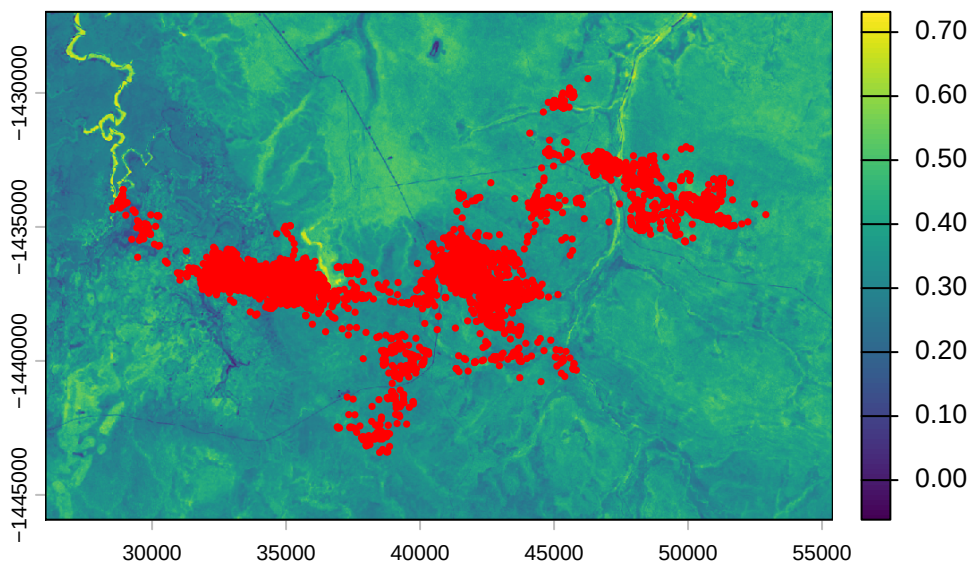


```
# create a buffer around the buffalo 2005 data
buffalo_clean_2005_sf <- buffalo_clean_2005 %>%
  st_as_sf(coords = c("lon", "lat"), crs = 4326) %>%
  st_transform(crs = 3112)

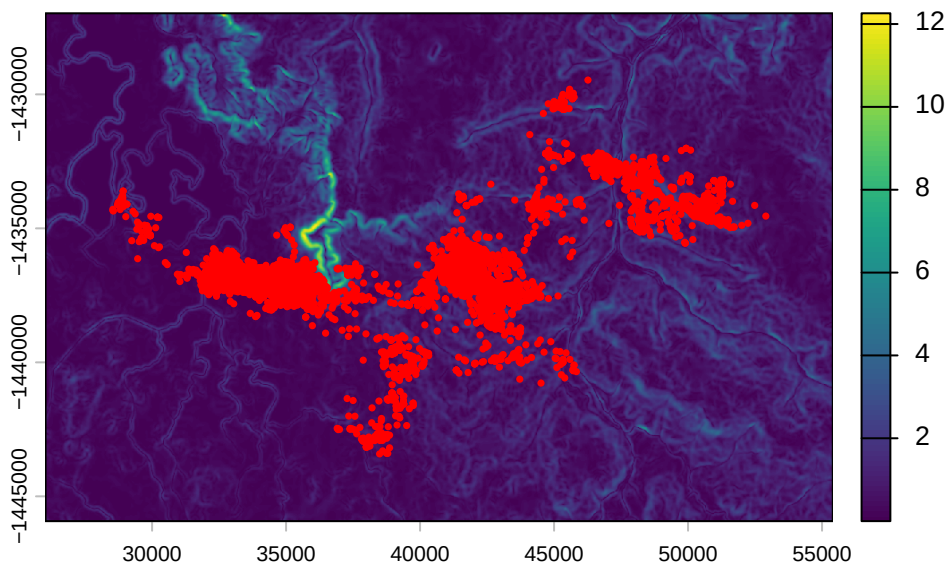
buffalo_clean_2005_extent <- buffalo_clean_2005_sf %>%
  st_buffer(dist = buffer_distance) %>%
  st_bbox() %>%
  st_as_sfc()

# crop the rasters to the extent of the buffalo 2005 data with the buffer distance
ndvi_2005 <- terra::crop(ndvi_average, buffalo_clean_2005_extent)
slope_2005 <- terra::crop(slope, buffalo_clean_2005_extent)

plot(ndvi_2005)
points(buffalo_clean_2005_sf$geometry, pch = 16, cex = 0.5, col = "red")
```

```
plot(slope_2005)
points(buffalo_clean_2005_sf$geometry, pch = 16, cex = 0.5, col = "red")
```



```
# write the rasters to file
terra::writeRaster(ndvi_2005, "mapping/cropped rasters/ndvi_2005.tif", overwrite = T)
terra::writeRaster(slope_2005, "mapping/cropped rasters/slope_2005.tif", overwrite = T)
```

Set up the spatial extent of the local covariates

We calculate the number of cells in each axis, and set the lag between locations. Typically this will just be 1, as we want the next-step to be the target. However, it may be possible

to improve model training by using different lags and including the time difference between locations as a covariate. I expect that this would help to predict across different time scales and not be restricted to the time scale that the data was collected at.

```
# calculate the number of cells in each axis
nxn_cells <- buffer*2/res
print(paste0("Number of cells in each axis: ", nxn_cells))
```

```
[1] "Number of cells in each axis: 101"
```

```
# hourly lag - to set larger time differences between locations
hourly_lag <- 1
```

Loop over each individual and save the local rasters

This is the main function of the script. It loops over each individual and calculates step level information, such as the location and time of the next step (x2, y2, t2), step length, turning angle, and different time components.

It also determines where to crop the local layers by subtracting and adding the buffer distance from the location of each step. It then crops out the local layers for each step and saves them as a list entry into the data frame.

It then creates a raster stack (with the number of layers equal to the number of steps) for each covariate and saves it as a tif.

For each individual, this will therefore create a csv containing the step level information, and a tif for each covariate containing the local layers for each step.

It takes quite a while to run, around 1 hour per individual in our case, so we will illustrate the function with 10 steps per individual. Uncomment the line specified in the loop to use the full dataset.

```
# for(i in 1:length(buffalo_ids)) {
for(i in 1:1) { # select just the first ID

  buffalo_data <- buffalo_all %>% filter(id == buffalo_ids[i])
  # all data for that individual
  buffalo_data <- buffalo_data %>% arrange(t_)

  # to use a subset of the data for that individual for testing
  # COMMENT THIS LINE OUT TO USE THE FULL DATASET
  buffalo_data <- buffalo_data %>% arrange(t_) |> slice(1:100)
```

```

n_samples <- nrow(buffalo_data)

tic()

buffalo_data_covs <- buffalo_data %>% mutate(

  x1_ = x_,
  y1_ = y_,
  x2_ = lead(x1_, n = hourly_lag, default = NA),
  y2_ = lead(y1_, n = hourly_lag, default = NA),
  x2_cent = x2_ - x1_,
  y2_cent = y2_ - y1_,
  t2_ = lead(t_, n = hourly_lag, default = NA),
  t_diff = round(difftime(t2_, t_, units = "hours"),0),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t1 = round(lubridate::hour(t_) + lubridate::minute(t_)/60,2),
  yday_t1 = round(lubridate::yday(t_) + lubridate::hour(t_)/24,2),

  hour_t1_sin1 = sin(2*pi*hour_t1/24),
  hour_t1_cos1 = cos(2*pi*hour_t1/24),
  hour_t1_sin2 = sin(4*pi*hour_t1/24),
  hour_t1_cos2 = cos(4*pi*hour_t1/24),
  hour_t1_sin3 = sin(6*pi*hour_t1/24),
  hour_t1_cos3 = cos(6*pi*hour_t1/24),

  yday_t1_sin1 = sin(2*pi*yday_t1/365.25),
  yday_t1_cos1 = cos(2*pi*yday_t1/365.25),
  yday_t1_sin2 = sin(4*pi*yday_t1/365.25),
  yday_t1_cos2 = cos(4*pi*yday_t1/365.25),
  yday_t1_sin3 = sin(6*pi*yday_t1/365.25),
  yday_t1_cos3 = cos(6*pi*yday_t1/365.25),

  # add the minutes as a decimal to the hour, and the hour as a decimal to the day
  hour_t2 = round(lubridate::hour(t2_) + lubridate::minute(t2_)/60,2),
  hour_t2_sin = sin(2*pi*hour_t2/24),
  hour_t2_cos = cos(2*pi*hour_t2/24),

  yday_t2 = round(lubridate::yday(t2_) + lubridate::hour(t2_)/24,2),
  yday_t2_sin = sin(2*pi*yday_t2/365.25),
  yday_t2_cos = cos(2*pi*yday_t2/365.25),

  sl = c(sqrt(diff(y_)^2 + diff(x_)^2), NA),
  log_sl = log(sl),

```

```

bearing = c(atan2(diff(y_), diff(x_)), NA),
bearing_sin = sin(bearing),
bearing_cos = cos(bearing),
ta = c(NA, ifelse(
  diff(bearing) > pi, diff(bearing)-(2*pi), ifelse(
    diff(bearing) < -pi, diff(bearing)+(2*pi), diff(bearing))),
cos_ta = cos(ta),

# extent for cropping the spatial covariates
x_min = x_ - buffer,
x_max = x_ + buffer,
y_min = y_ - buffer,
y_max = y_ + buffer,

# crop out and store the local covariates centered on the animal's location
# with an extent set in the previous chunk
) %>% rowwise() %>% mutate(

# the extent needs to be the same for saving the stack of rasters
extent_00centre = list(ext(x_min - x_, x_max - x_, y_min - y_, y_max - y_)),

# NDVI
ndvi_index = which.min(abs(difftime(t_, terra::time(ndvi_projected)))),
ndvi_cent = list({
  ndvi_cent = crop(ndvi_projected[[ndvi_index]], ext(x_min, x_max, y_min, y_max))
  ext(ndvi_cent) <- extent_00centre
  ndvi_cent
}),

# herbaceous vegetation
veg_herby_cent = list({
  veg_herby_cent = crop(veg_herby, ext(x_min, x_max, y_min, y_max))
  ext(veg_herby_cent) <- extent_00centre
  veg_herby_cent
}),

# canopy cover
canopy_cover_cent = list({
  canopy_cover_cent = crop(canopy_cover, ext(x_min, x_max, y_min, y_max))
  ext(canopy_cover_cent) <- extent_00centre
  canopy_cover_cent
}),

```



```

# slope
slope_cent = list({
  slope_cent <- crop(slope, ext(x_min, x_max, y_min, y_max))
  ext(slope_cent) <- extent_00centre
  slope_cent
}),

# presence (next-step)
pres_template = list({
  pres_template <- crop(slope, ext(x_min, x_max, y_min, y_max))
  pres_template
}),

# Strict within cell method
points_vect_cent_x2_within_cell = list(terra::vect(cbind(x2_, y2_), type = "points", crs
pres_cent_x2_within_cell = list({
  pres_cent_x2_within_cell <- rasterize(points_vect_cent_x2_within_cell, pres_template,
  ext(pres_cent_x2_within_cell) <- extent_00centre
  pres_cent_x2_within_cell
})

) %>% ungroup() # to remove the 'rowwise' class

toc()

# remove steps that fall outside of the local spatial extent
buffalo_data_covs <- buffalo_data_covs %>%
  filter(x2_cent > -buffer & x2_cent < buffer & y2_cent > -buffer & y2_cent < buffer) %>%
  drop_na(ta)

# remove the columns that are no longer needed
buffalo_data_df <- buffalo_data_covs %>%
  dplyr::select(
    -extent_00centre,
    -ndvi_cent,
    -veg_herby_cent,
    -canopy_cover_cent,
    -slope_cent,
    -pres_template,
    -points_vect_cent_x2_within_cell,
    -pres_cent_x2_within_cell)

# save the data

```

```

write_csv(buffalo_data_df, paste0("buffalo_local_data_id/buffalo_temporal_cont_", buffalo_
      "_data_df_lag_", hourly_lag, "hr_n", n_samples, ".csv"))

# saving the raster objects
rast(buffalo_data_covs$ndvi_cent) %>%
  writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_ndvi_cent",
    nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
    overwrite = T)

rast(buffalo_data_covs$veg_herby_cent) %>%
  writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_herby_cent",
    nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
    overwrite = T)

rast(buffalo_data_covs$canopy_cover_cent) %>%
  writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_canopy_cent",
    nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
    overwrite = T)

rast(buffalo_data_covs$slope_cent) %>%
  writeRaster(paste0("buffalo_local_layers_id/buffalo_", buffalo_ids[i], "_slope_cent",
    nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
    overwrite = T)

rast(buffalo_data_covs$pres_cent_x2_within_cell) %>%
  writeRaster(paste0("buffalo_local_layers_id/within_cell_buffalo_", buffalo_ids[i], "_pre
    nxn_cells, "x", nxn_cells, "_lag_", hourly_lag, "hr_n", n_samples, ".
    overwrite = T)
}

```

6.304 sec elapsed

Check the outputs

```
head(buffalo_data_df)
```

```
# A tibble: 6 x 44
```

	x_	y_	t_		id	x1_	y1_	x2_	y2_	x2_cent
	<dbl>	<dbl>	<dtm>		<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	41969.	-1.44e6	2018-07-25 11:04:23	2005	41969.	-1.44e6	41922.	-1.44e6	-	
47.8										

```

2 41922. -1.44e6 2018-07-25 12:04:39 2005 41922. -1.44e6 41779. -1.44e6 -
142.
3 41779. -1.44e6 2018-07-25 13:04:17 2005 41779. -1.44e6 41841. -1.44e6 61.8
4 41841. -1.44e6 2018-07-25 14:04:39 2005 41841. -1.44e6 41655. -1.44e6 -
186.
5 41655. -1.44e6 2018-07-25 15:04:27 2005 41655. -1.44e6 41619. -1.44e6 -
36.8
6 41619. -1.44e6 2018-07-25 16:04:24 2005 41619. -1.44e6 41688. -1.44e6 69.8
# i 35 more variables: y2_cent <dbl>, t2_ <dtm>, t_diff <drtn>, hour_t1 <dbl>,
# yday_t1 <dbl>, hour_t1_sin1 <dbl>, hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>,
# hour_t1_cos2 <dbl>, hour_t1_sin3 <dbl>, hour_t1_cos3 <dbl>,
# yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
# yday_t1_cos2 <dbl>, yday_t1_sin3 <dbl>, yday_t1_cos3 <dbl>, hour_t2 <dbl>,
# hour_t2_sin <dbl>, hour_t2_cos <dbl>, yday_t2 <dbl>, yday_t2_sin <dbl>,
# yday_t2_cos <dbl>, sl <dbl>, log_sl <dbl>, bearing <dbl>, ...

```

```
head(buffalo_data_covs)
```

```

# A tibble: 6 x 52
      x_      y_ t_      id      x1_      y1_      x2_      y2_ x2_cent
  <dbl> <dbl> <dtm>    <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
1 41969. -1.44e6 2018-07-25 11:04:23 2005 41969. -1.44e6 41922. -1.44e6 -
47.8
2 41922. -1.44e6 2018-07-25 12:04:39 2005 41922. -1.44e6 41779. -1.44e6 -
142.
3 41779. -1.44e6 2018-07-25 13:04:17 2005 41779. -1.44e6 41841. -1.44e6 61.8
4 41841. -1.44e6 2018-07-25 14:04:39 2005 41841. -1.44e6 41655. -1.44e6 -
186.
5 41655. -1.44e6 2018-07-25 15:04:27 2005 41655. -1.44e6 41619. -1.44e6 -
36.8
6 41619. -1.44e6 2018-07-25 16:04:24 2005 41619. -1.44e6 41688. -1.44e6 69.8
# i 43 more variables: y2_cent <dbl>, t2_ <dtm>, t_diff <drtn>, hour_t1 <dbl>,
# yday_t1 <dbl>, hour_t1_sin1 <dbl>, hour_t1_cos1 <dbl>, hour_t1_sin2 <dbl>,
# hour_t1_cos2 <dbl>, hour_t1_sin3 <dbl>, hour_t1_cos3 <dbl>,
# yday_t1_sin1 <dbl>, yday_t1_cos1 <dbl>, yday_t1_sin2 <dbl>,
# yday_t1_cos2 <dbl>, yday_t1_sin3 <dbl>, yday_t1_cos3 <dbl>, hour_t2 <dbl>,
# hour_t2_sin <dbl>, hour_t2_cos <dbl>, yday_t2 <dbl>, yday_t2_sin <dbl>,
# yday_t2_cos <dbl>, sl <dbl>, log_sl <dbl>, bearing <dbl>, ...

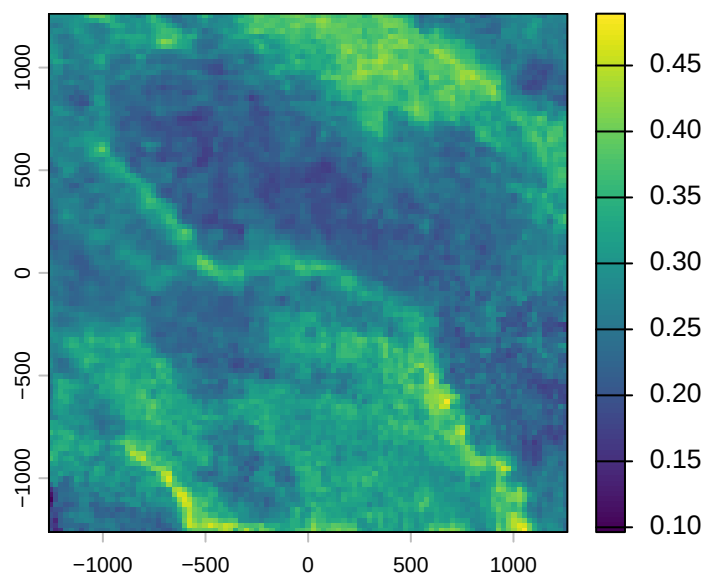
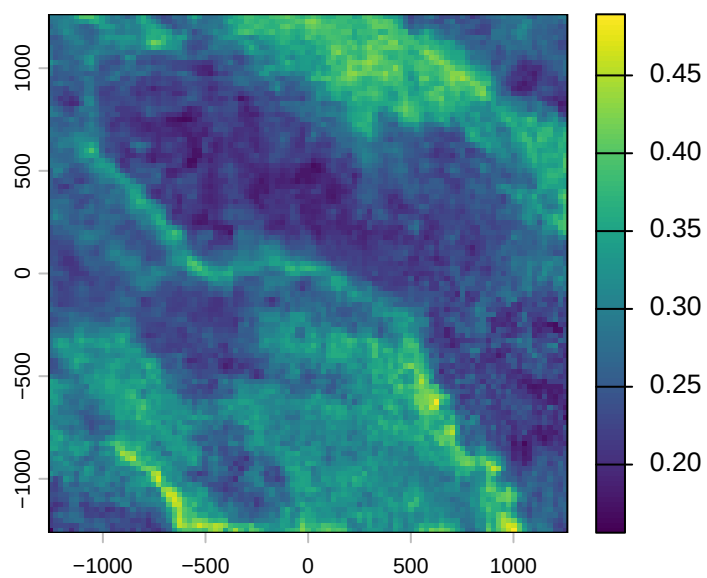
```

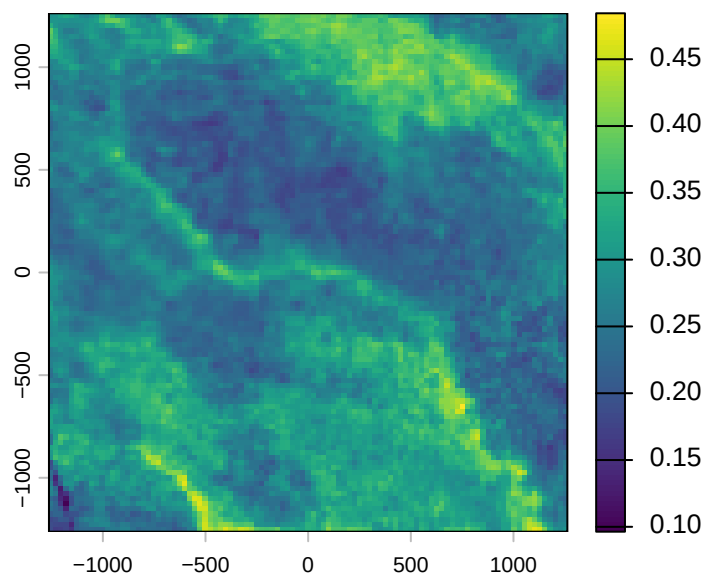
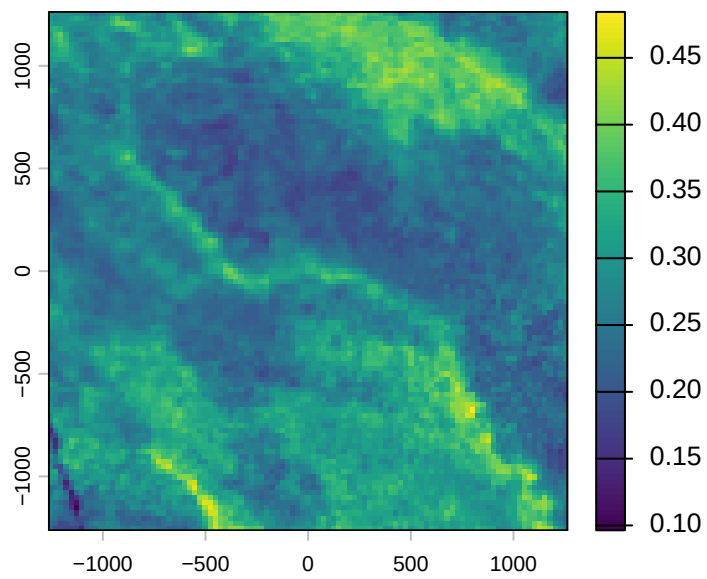
```
which(is.na(buffalo_data_covs$ta))
```

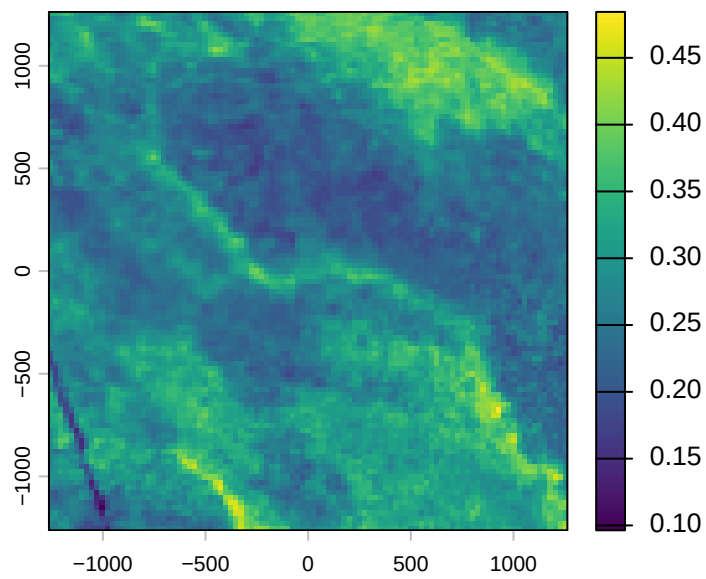
```
integer(0)
```

Plot a subset of the local covariates

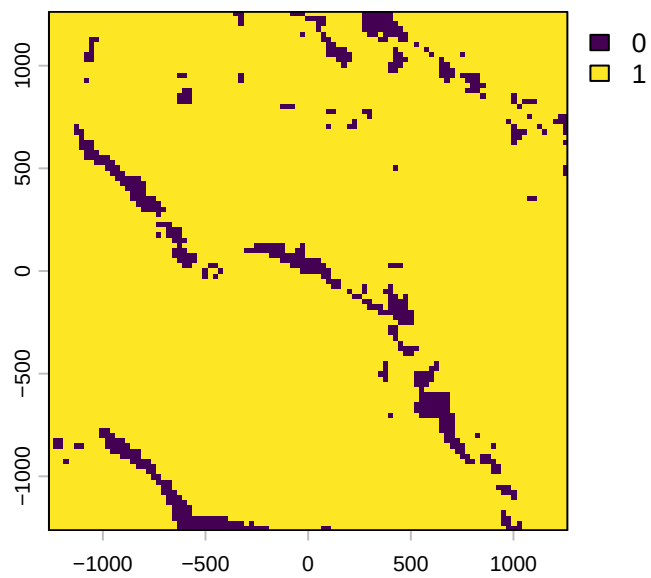
```
n_plots <- 5  
  
# NDVI  
walk(buffalo_data_covs$ndvi_cent[1:n_plots], terra::plot)
```

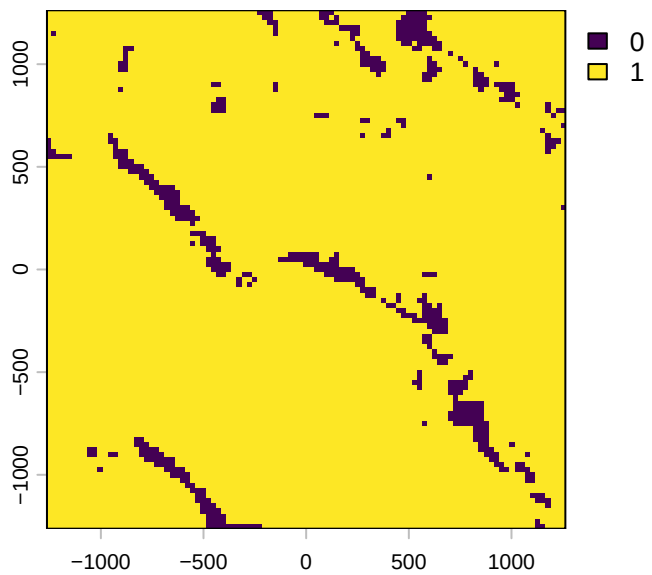
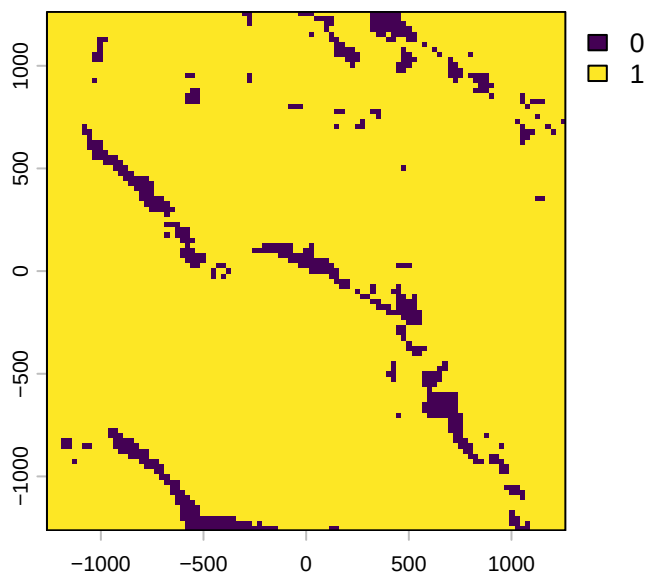


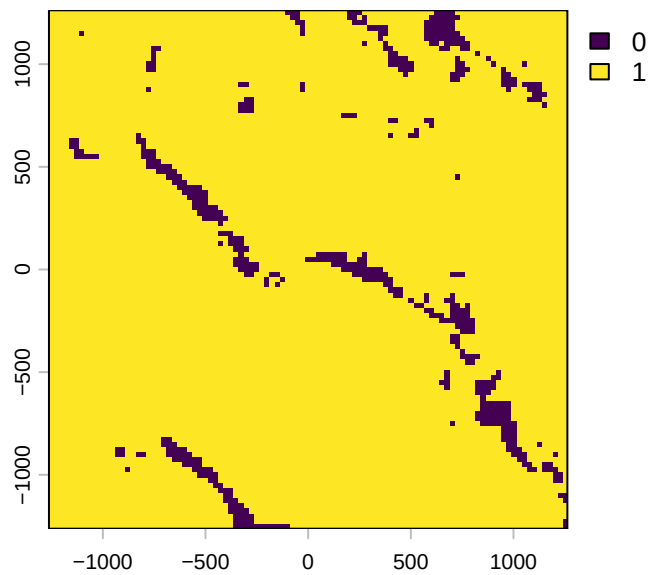
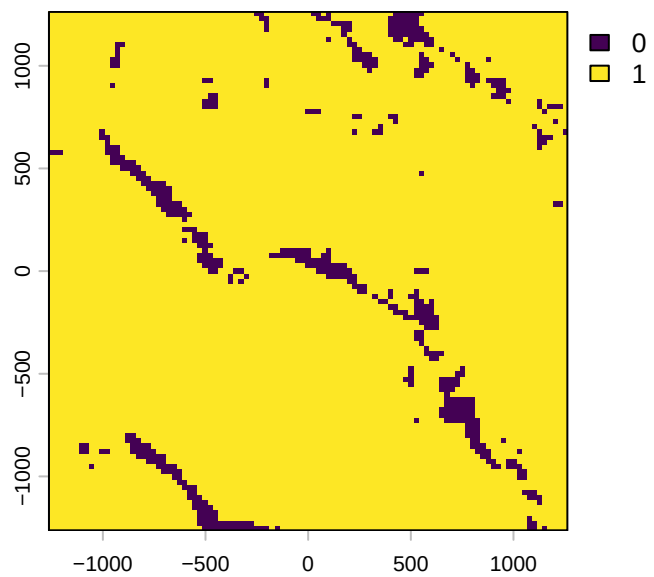




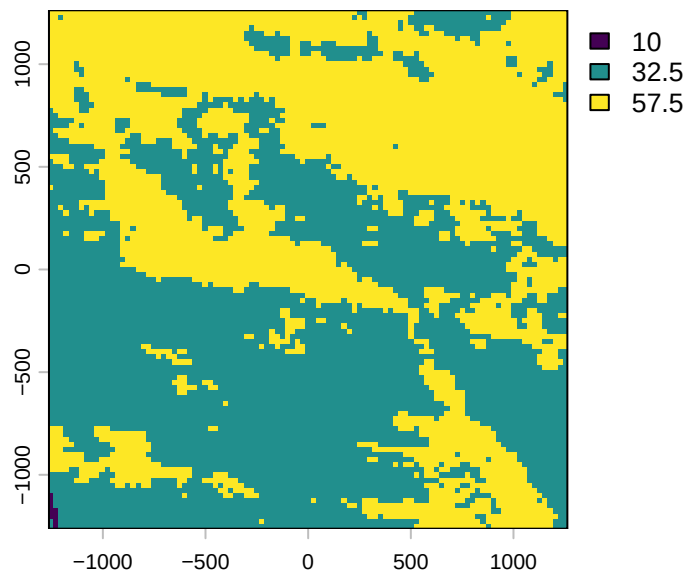
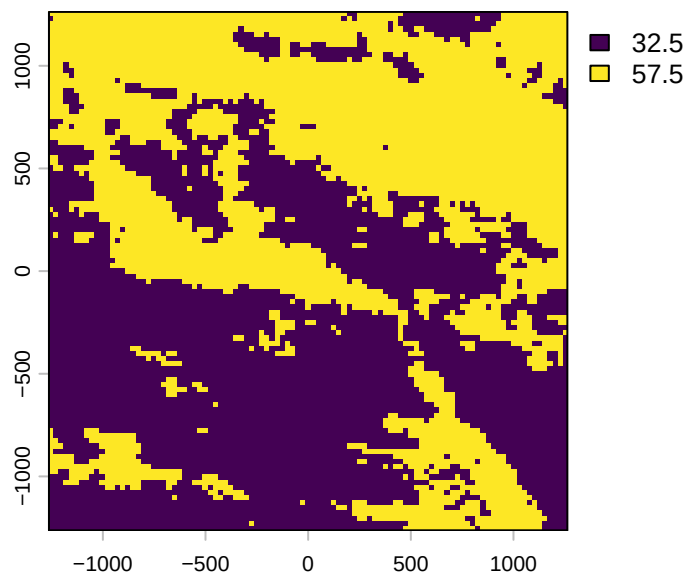
```
# Herbaceous vegetation
walk(buffalo_data_covs$veg_herby_cent[1:n_plots], terra::plot)
```

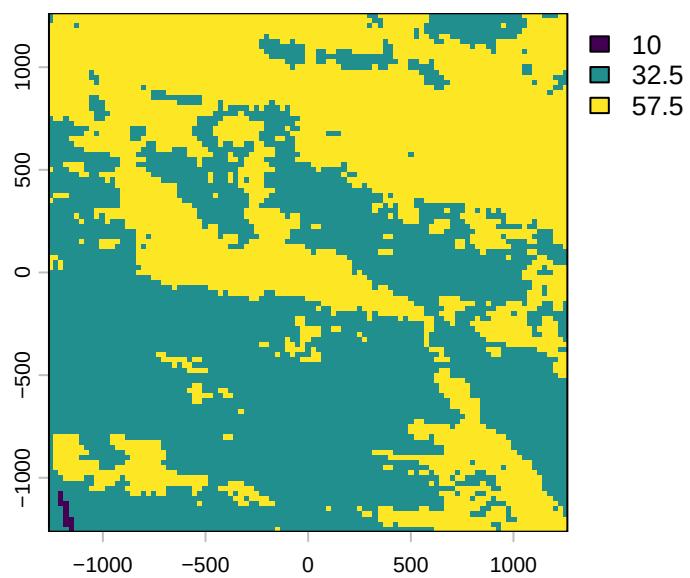
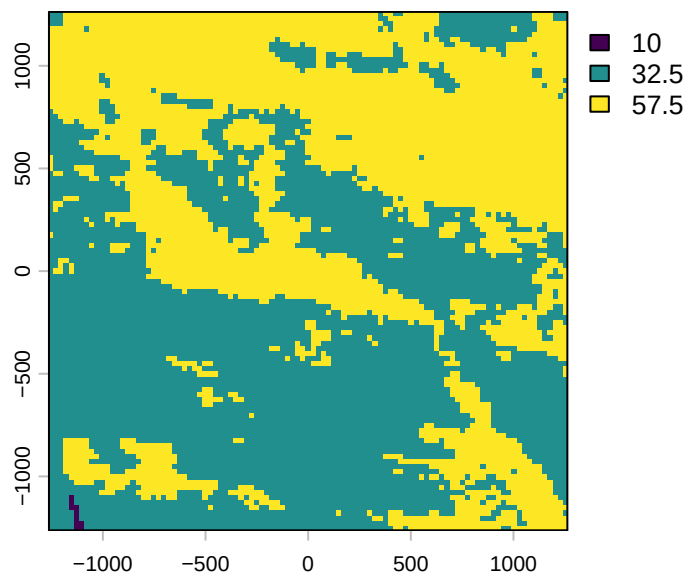


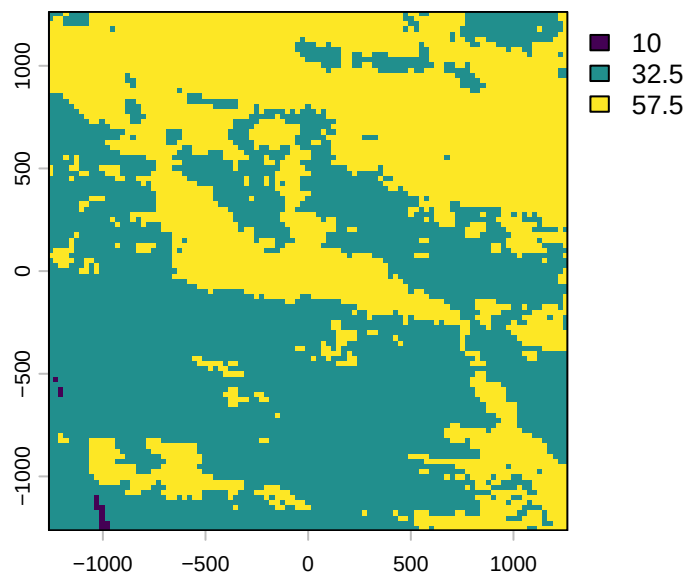




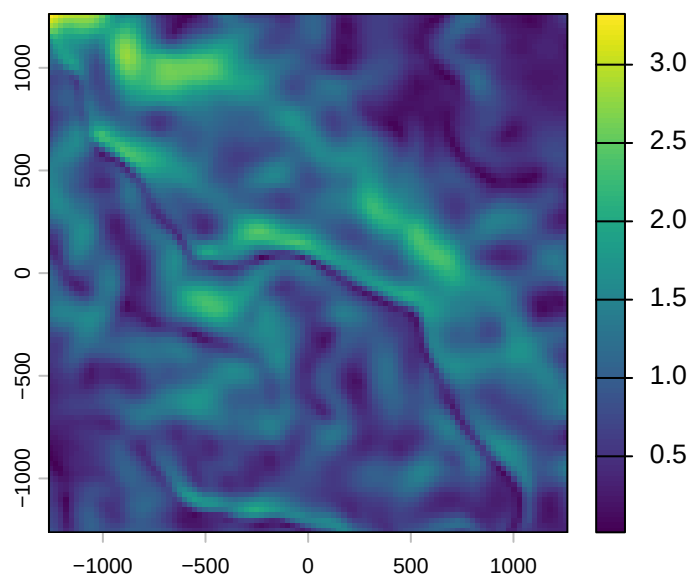
```
# Canopy cover
walk(buffalo_data_covs$canopy_cover_cent[1:n_plots], terra::plot)
```

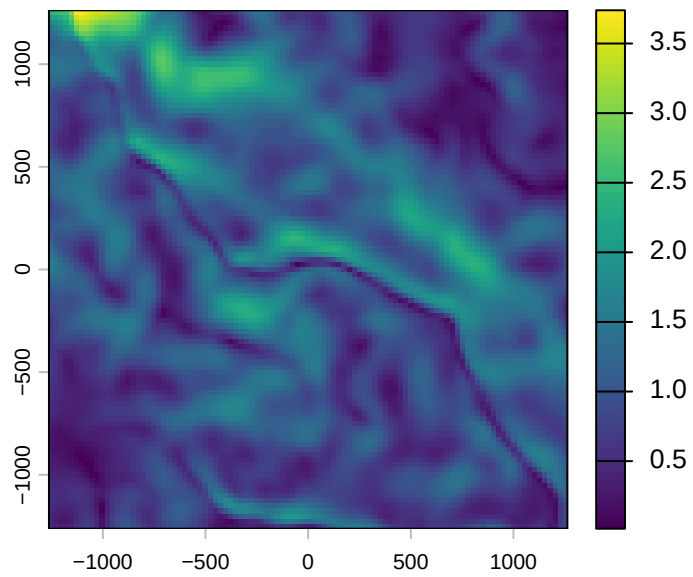
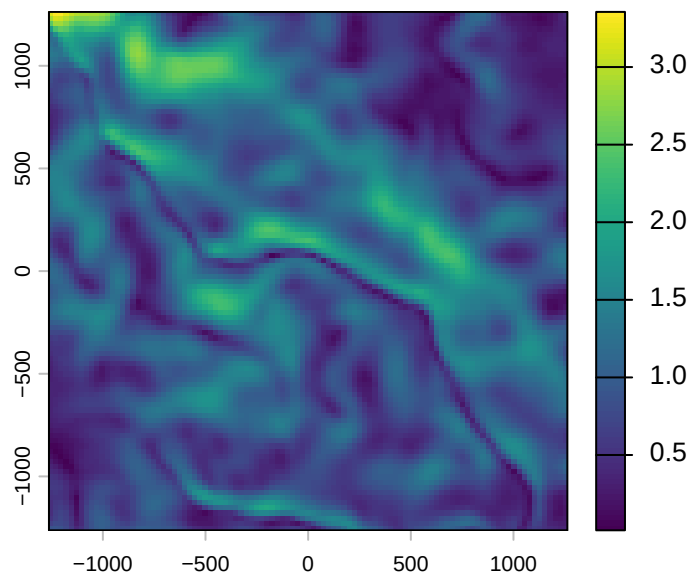



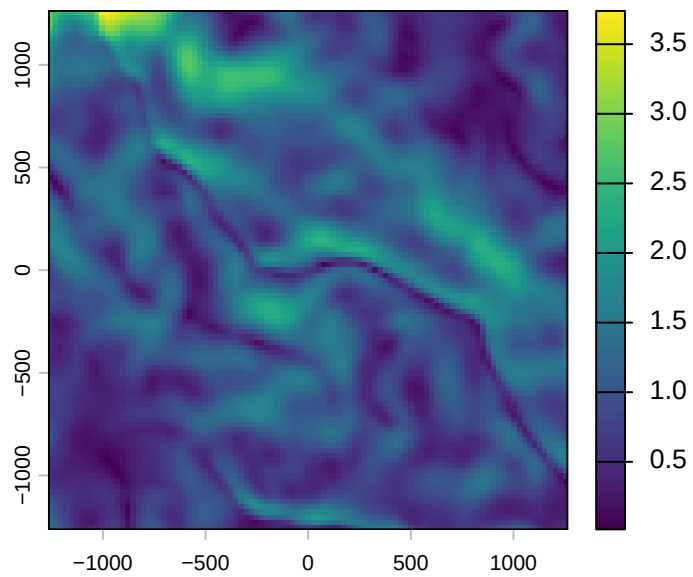
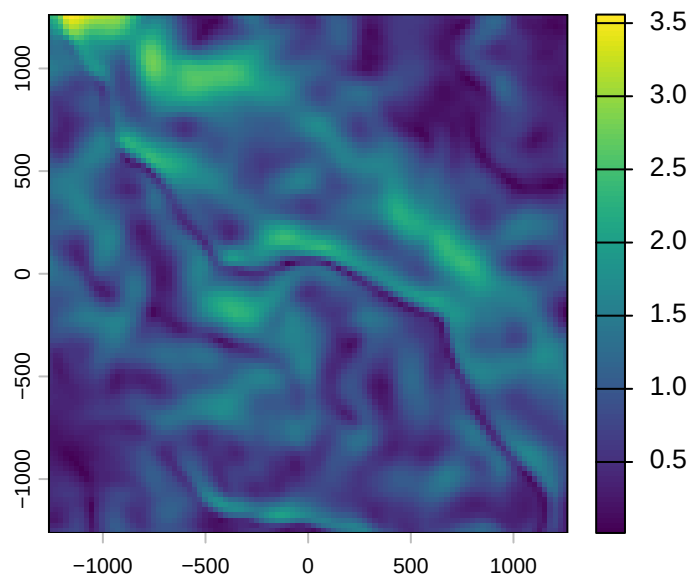




```
# Slope
walk(buffalo_data_covs$slope_cent[1:n_plots], terra::plot)
```



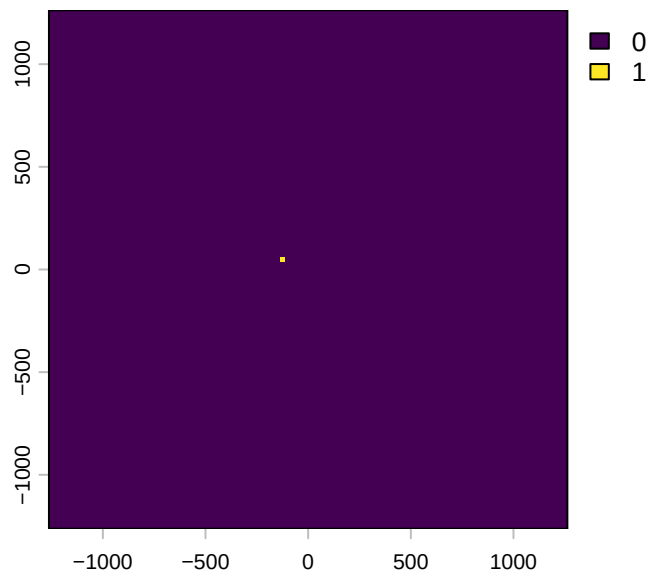
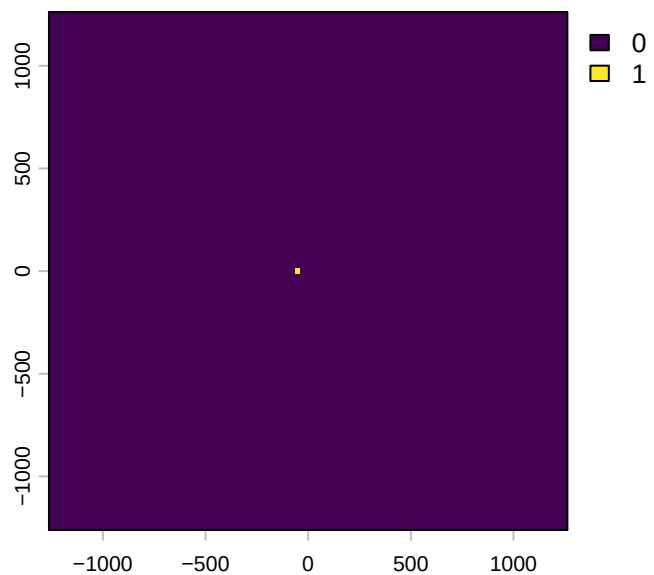


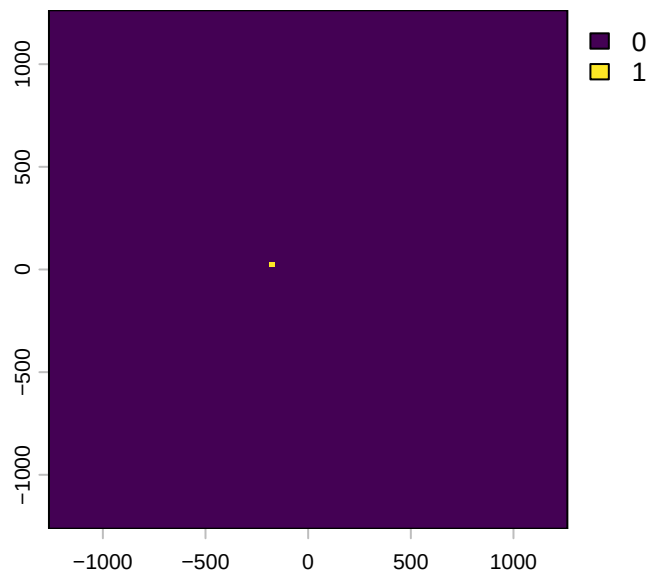
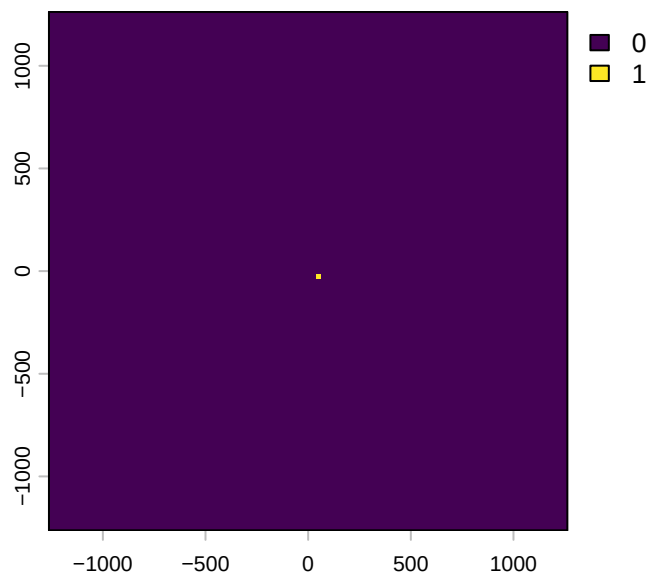


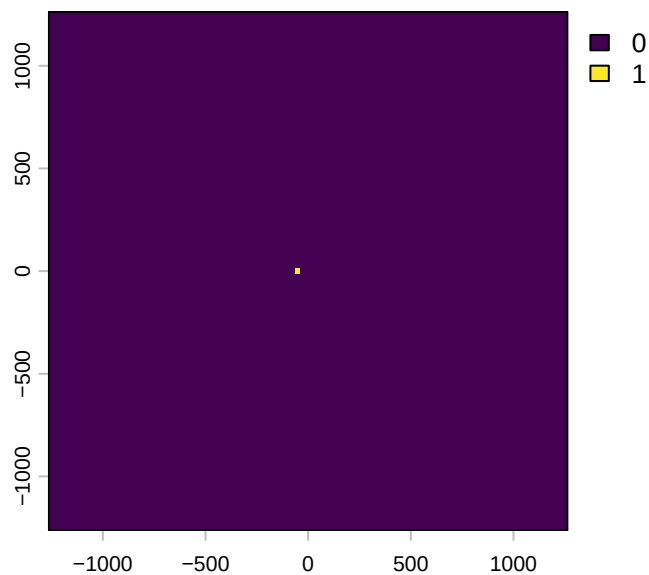
```
# Target (location of the next step)
walk(buffalo_data_covs$pres_cent_x1[1:n_plots], terra::plot)
```

Warning: Unknown or uninitialised column: `pres\_cent\_x1`.

```
# Target (location of the next step)
walk(buffalo_data_covs$pres_cent_x2_within_cell[1:n_plots], terra::plot)
```







To save the object with all of the local covariates

```
# saveRDS(buffalo_data_covs, paste0("buffalo_data_id_covs_", Sys.Date(), ".rds"))
```

To save some plots

```
# for(i in 1:n_plots) {
#   png(filename = paste0("ndvi_cent_", i, ".png"),
#     width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$ndvi_cent[[i]])
#   dev.off()
# }
#
# # for the pres layers
# for(i in 1:n_plots) {
#
#   # change the 0 values to NA
#   layer <- buffalo_data_covs$pres_cent[[i]]
#   layer[layer == 0] <- 0.5
#
#   ndvi_layer <- buffalo_data_covs$ndvi_cent[[i]]
#   new_layer <- layer*ndvi_layer
#
#   # save the plot
```



```

#   png(filename = paste0("pres_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(new_layer)
#   dev.off()
# }
#
#
# # other plots
#
# n_plots <- 1
#
# for(i in 1:n_plots) {
#   png(filename = paste0("veg_herby_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$veg_herby_cent[[i]])
#   dev.off()
# }
#
# for(i in 1:n_plots) {
#   png(filename = paste0("canopy_cover_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$canopy_cover_cent[[i]])
#   dev.off()
# }
#
# for(i in 1:n_plots) {
#   png(filename = paste0("slope_cent_", i, ".png"),
#       width = 150, height = 150, units = "mm", res = 600)
#   terra::plot(buffalo_data_covs$slope_cent[[i]])
#   dev.off()
# }

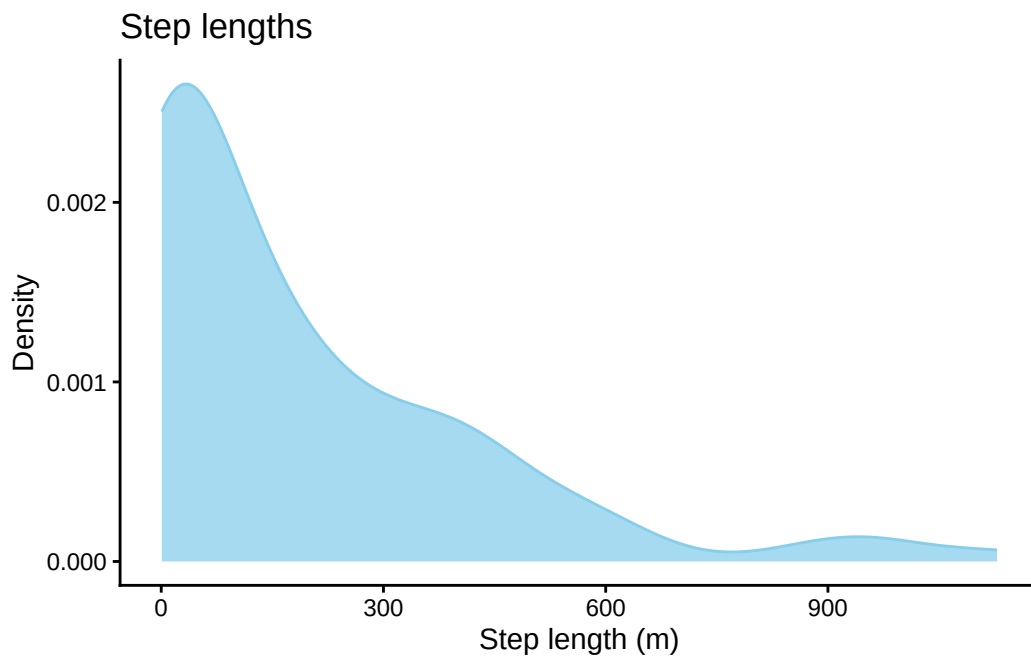
```

Plot the step lengths and turning angles

```

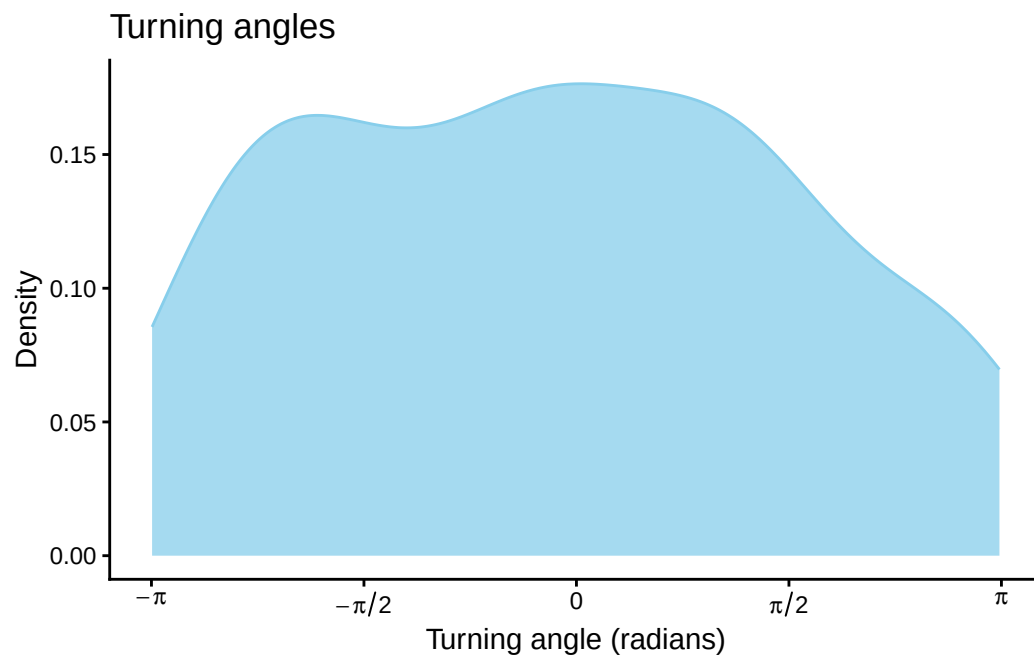
# plot step lengths
buffalo_data_covs %>% ggplot() +
  geom_density(aes(x = sl),
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  scale_x_continuous("Step length (m)") +
  scale_y_continuous("Density") +
  ggtitle("Step lengths") +
  theme_classic()

```



```
# ggsave("outputs/step_length.png", height = 60, width = 120, units = "mm", dpi = 600)

# plot turning angles
buffalo_data_covs %>% ggplot() +
  geom_density(aes(x = ta),
               fill = "skyblue", colour = "skyblue", alpha = 0.75) +
  scale_x_continuous("Turning angle (radians)",
                     breaks = c(-pi, -pi/2, 0, pi/2, pi),
                     labels = c(expression(-pi), expression(-pi/2), "0", expression(pi/2), expression(pi))
  ) +
  scale_y_continuous("Density") +
  ggtitle("Turning angles") +
  theme_classic()
```



```
# ggsave("outputs/turning_angle.png", height = 60, width = 120, units = "mm", dpi = 600)
```